

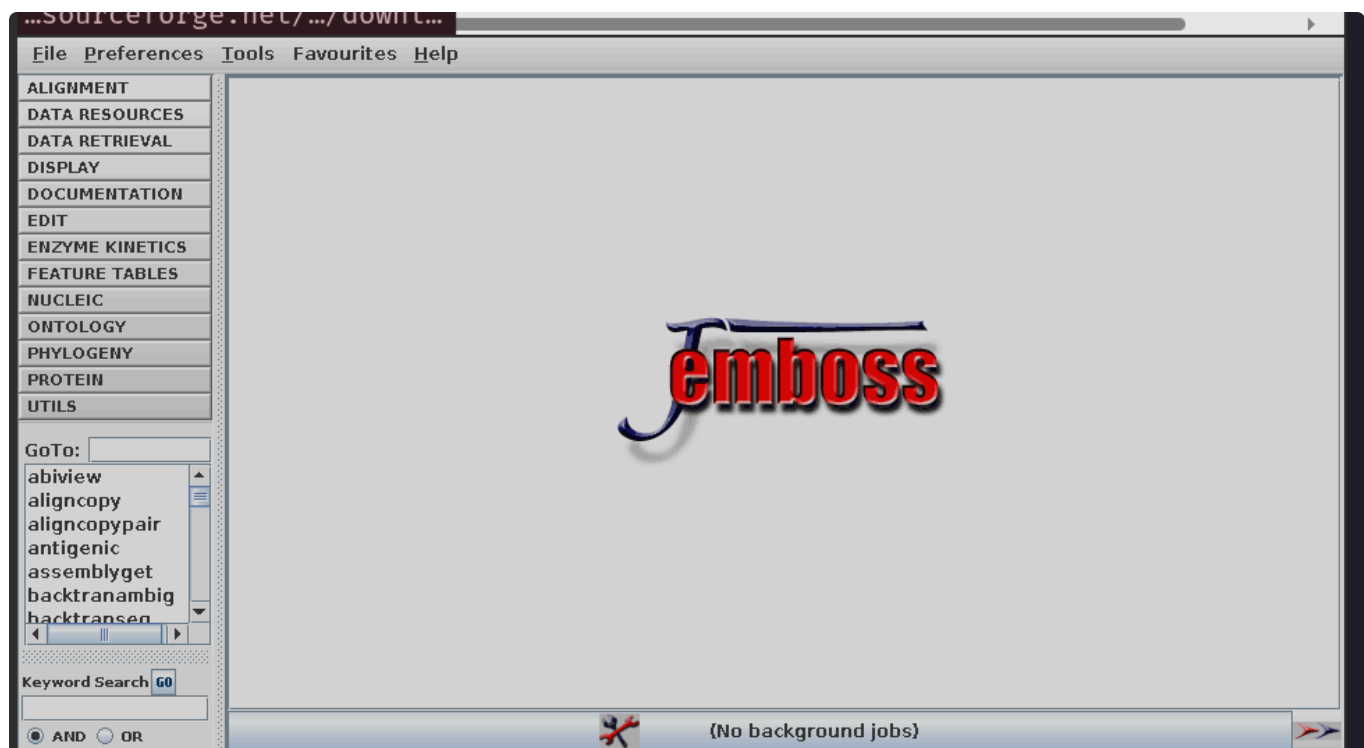
Solution -1

all csvs and python file are attached here

https://github.com/specapoorv/BT3040_BioInfo_Lab_assignments/tree/main/assignment_1

1. Download the EMBOSS package (<http://emboss.sourceforge.net/download/>) and copy to your Windows system. In case of **Linux** use the command: `sudo apt-get install jemboss`. For Mac users, use the online tool links given in the instructions doc.

solution :



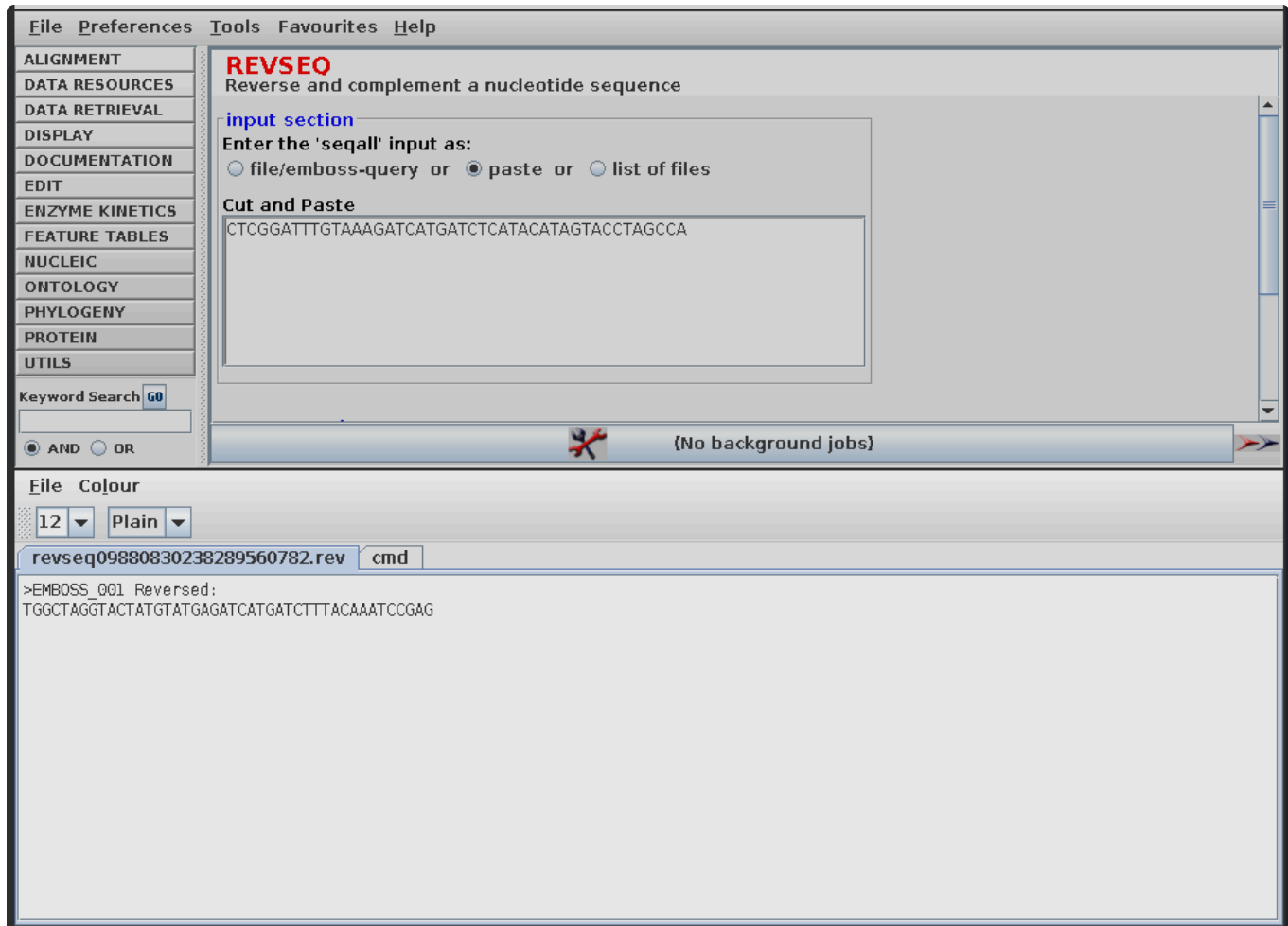
downloaded 👍

Q2 Using EMBOSS, find the complementary strand for the sequence:
CTCGGATTTGTAAAGATCATGATCTCATACATAGTA
CCTAGCCA

using terminal

```
> revseq -sequence asis:CTCGGATTTGTAAAGATCATGATCTCATACATAGTACCTAGCCA -outseq stdout -noreverse -complement
Reverse and complement a nucleotide sequence
>asis
GAGCCTAAACATTTCTAGTACTAGAGTATGTATCATGGATCGGT
~ > |
```

using GUI



Q3 Write a program to find the complementary strand for the sequence given in Q2.

```
#seq : str = input() #should be string can be typecasted

seq = "CTCGGATTTGTAAAGATCATGATCTCATACATAGTACCTAGCCA"
seq.upper()

complement_map = {'A':'T', 'T':'A', "G":"C", 'C':'G'}

#o(1) comprehension since hashmap
```

```
complementary_strand = "".join([complement_map[base] for  
base in seq])
```

```
print("given strand is", seq)  
print("its complement is", complementary_strand)
```

Q.4

The screenshot shows the TRANSEQ web interface. The left sidebar contains a menu with options: ALIGNMENT, DATA RESOURCES, DATA RETRIEVAL, DISPLAY, DOCUMENTATION, EDIT, ENZYME KINETICS, FEATURE TABLES, NUCLEIC, ONTOLOGY, PHYLOGENY, PROTEIN, and UTILS. The 'UTILS' section is expanded, showing a list of tools: GoTo: trans, rexger, textsearch, tfm, tfscan, tmap, tranalign, and transeq (highlighted). Below the list is a 'Keyword Search' field with a 'GO' button. The main area is titled 'TRANSEQ' and 'Translate nucleic acid sequences'. It has an 'input section' with a text area containing the sequence: GACATTGTGAACAGTAAAAAGTCCATGCAATGCGCAAGGAGCAGAAGAGGAAGCAGGCTCTCCCCTGGACTACTCTCTCTGCCCATCGACAAGCATGAGCCTGAATTTGGTCCATGG. Below the text area is a 'Cut and Paste' button. The 'output section' has a text area for 'Output Sequence Name' and a button for 'Output Sequence Options'. The 'Execution mode' is set to 'interactive', with a 'GO' button and an 'Advanced Options' button. At the bottom, there is a status bar showing 'No background jobs'. The bottom panel shows the command 'transeq017480133008298899185.pep' and the output sequence: >_1
DIVNSKKVHAMRKEQKRKQGKQRSHGSPHDYSPLPIDKHEPEFGPCRRLDG.

TRANSEQ
Translate nucleic acid sequences

interactive Advanced Options

additional section

-1
-2
-3
Reverse three frames
All six frames

Frame(s) to translate (min:1 max:6 default:1)

Standard Code to use (min:1 max:1 default:0)

Regions to translate (eg: 4-57,78-94)

Keyword Search

AND OR

{No background jobs}

File Colour

12 Plain

transeq0646292218600613325.pep cmd

```
> 1
DTVNSKKVHAMRKEQKRKQGGKQSMGSPMDYSPLPIDKHEPEFGPCRRLDG
> 2
TL*TVKKSMQCARSRRGRASSAPWALPWTTLLCPSTSMNLVHAEENWMG
> 3
HCEQ*KSPCNAQGAEEEEAGQAALHGLSHGLLSSAHRQA*A*IWSMQKTGWX
> 4
PTQFSSAWTKFRLMLVDGQRRVHGRAHGALLALLPLLLLAHMDFFT VHNV
> 5
HPVFFCMDQIQAHACRWAEESPWESPWSAACPASSAPCALHGLFYCSQCX
> 6
PSSFLLHGPNSGCLSMGRGE*SMGEPMERCLPCFLFCSLRIAWTFLLFTMS
```

Answer : 4th reading frame

Q.5 Write a program to translate the given DNA sequence (refer Q4) to the protein sequence.

```
def translate_dna(dna_seq):

    codon_table = {
        'ATA': 'I', 'ATC': 'I', 'ATT': 'I', 'ATG': 'M',
        'ACA': 'T', 'ACC': 'T', 'ACG': 'T', 'ACT': 'T',
        'AAC': 'N', 'AAT': 'N', 'AAA': 'K', 'AAG': 'K',
        'AGC': 'S', 'AGT': 'S', 'AGA': 'R', 'AGG': 'R',
        'CTA': 'L', 'CTC': 'L', 'CTG': 'L', 'CTT': 'L',
        'CCA': 'P', 'CCC': 'P', 'CCG': 'P', 'CCT': 'P',
        'CAC': 'H', 'CAT': 'H', 'CAA': 'Q', 'CAG': 'Q',
        'CGA': 'R', 'CGC': 'R', 'CGG': 'R', 'CGT': 'R',
        'GTA': 'V', 'GTC': 'V', 'GTG': 'V', 'GTT': 'V',
        'GCA': 'A', 'GCC': 'A', 'GCG': 'A', 'GCT': 'A',
        'GAC': 'D', 'GAT': 'D', 'GAA': 'E', 'GAG': 'E',
        'GGA': 'G', 'GGC': 'G', 'GGG': 'G', 'GGT': 'G',
```

```

        'TCA': 'S', 'TCC': 'S', 'TCG': 'S', 'TCT': 'S',
        'TTC': 'F', 'TTT': 'F', 'TTA': 'L', 'TTG': 'L',
        'TAC': 'Y', 'TAT': 'Y', 'TAA': '*', 'TAG': '*',
        'TGC': 'C', 'TGT': 'C', 'TGA': '*', 'TGG': 'W',
    }

    protein = ""

    for i in range(0, len(dna_seq) - 2, 3):
        codon = dna_seq[i:i+3]
        protein += codon_table.get(codon, "")

    return protein

dna_sequence =
"GACATTGTGAACAGTAAAAAAGTCCATGCAATGCGCAAGGAGCAGAAGAGGAAGCAG
GGCAAGCAGCGCTCCATGGGCTCTCCCATGGACTACTCTCCTCTGCCCATCGACAAGC
ATGAGCCTGAATTTGGTCCATGCAGAAGAAAACCTGGATGGG"
#or we can do input() as well
protein_sequence = translate_dna(dna_sequence)

print(f"DNA Sequence: {dna_sequence}")
print(f"Protein Sequence: {protein_sequence}")

```

Q.6

```

# The DNA sequence provided in Question 4
dna_sequence =
"GACATTGTGAACAGTAAAAAAGTCCATGCAATGCGCAAGGAGCAGAAGAGGAAGCAG
GGCAAGCAGCGCTCCATGGGCTCTCCCATGGACTACTCTCCTCTGCCCATCGACAAGC
ATGAGCCTGAATTTGGTCCATGCAGAAGAAAACCTGGATGGG"
search_strings = ['AAG', 'GTC', 'GAG', 'ACTA', 'ATAT']

for target in search_strings:

    print(f"Enter the string: {target}")
    positions = []
    index = dna_sequence.find(target)
    while index != -1:
        positions.append(str(index + 1))

```

```
        index = dna_sequence.find(target, index + 1)
    print(f"Total match: {len(positions)}")
    if positions:
        print(f"Position of match: {'',
        '.join(positions)}")
    else:
        print("Position of match: 0")
```

```
> python3 bs.py
Enter the string: AAG
Total match: 7
Position of match: 20, 37, 46, 52, 61, 112, 141
Enter the string: GTC
Total match: 2
Position of match: 22, 131
Enter the string: GAG
Total match: 3
Position of match: 40, 48, 118
Enter the string: ACTA
Total match: 1
Position of match: 89
Enter the string: ATAT
Total match: 0
Position of match: 0
~ > █
```

Q.7

File Preferences Tools Favourites Help

ALIGNMENT

DATA RESOURCES

DATA RETRIEVAL

DISPLAY

DOCUMENTATION

EDIT

ENZYME KINETICS

FEATURE TABLES

NUCLEIC

ONTOLOGY

PHYLOGENY

PROTEIN

UTILS

GoTo: dan

abiview

aligncopy

aligncoppair

antigenic

assemblyget

backtranambig

backtransen

Keyword Search

GO

AND

OR

DAN

Calculate nucleic acid melting temperature

input section

Enter the 'seqall' input as:

file/emboss-query

 or

paste

 or

list of files

Cut and Paste

GACATTGTGAACAGTAAAAAGTCCATGCAATGCGCAAGGAGCAGAAAGGGAAGCAGGCTCTCCCATGGACTACTCTCTCTGCCCCATCGACAAGCATGAGCCTGAATTGGTCCATGGG

LOAD SEQUENCE ATTRIBUTES

required section

20

Enter window size
(min:1 max:100 default:20)

1

Enter shift increment
(min:1 default:1)

50

Enter DNA concentration (nM)
(min:1. max:100000. default:50.)

50

Enter salt concentration (mM)

(No background jobs)

File Colour

12

Plain

dan013078456202101548748.dan

cmd

#####

Program: dan

Rundate: Tue 11 Aug 2026 17:58:12

Commandline: dan

-sequence /tmp/dan013078456202101548748.jembosstp

-window size 20

-shift increment 1

-dnaconc 50.0

-saltconc 50.0

-nopproduct

-nothermo

-norna

-noplot

-graph png

-rformat seqtable

-auto

Report_format: seqtable

Report_file: dan013078456202101548748.dan

#####

#=====

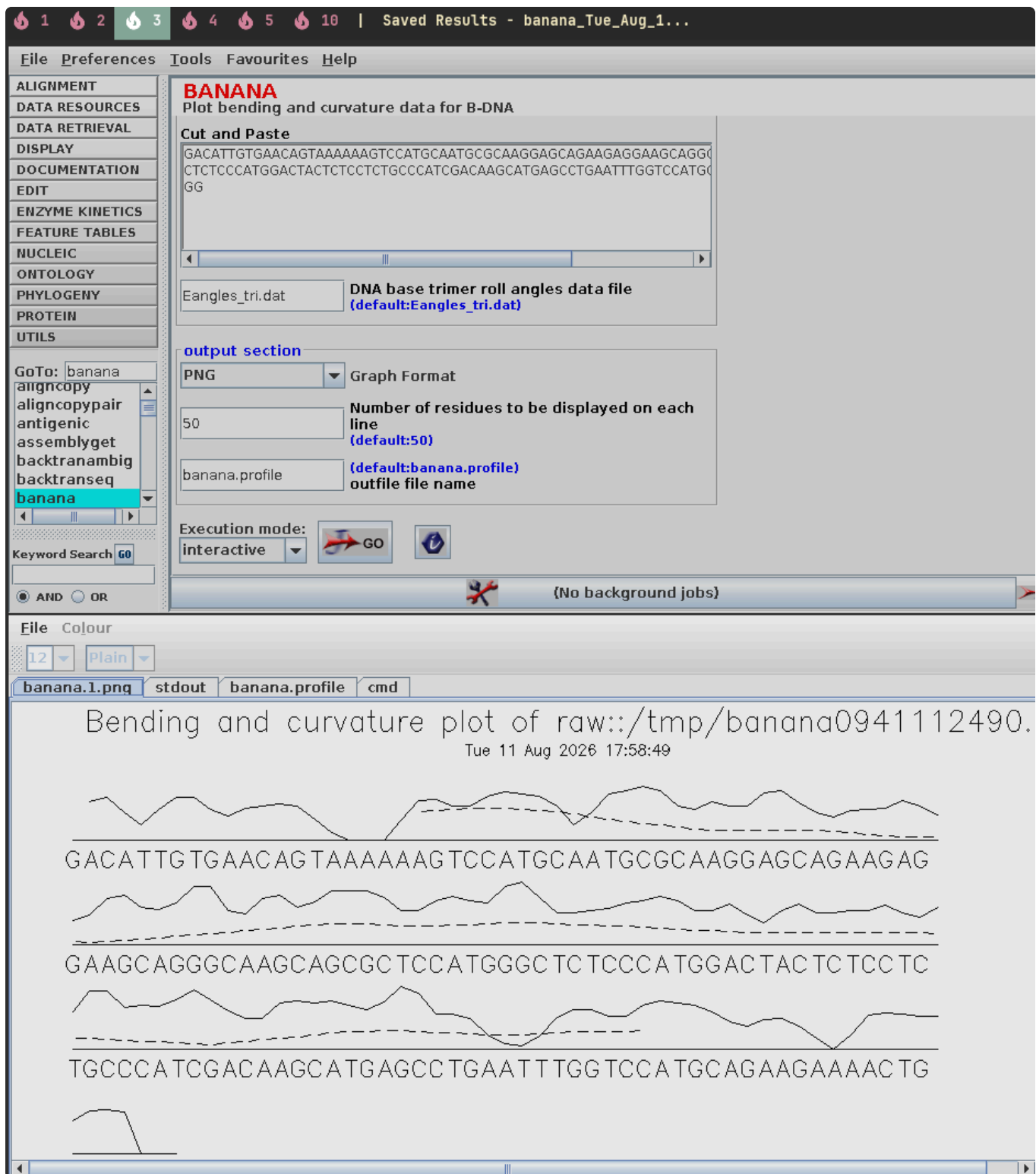
#

Sequence: from: 1 to: 156

HitCount: 137

#=====

Start	End	Strand	Tm	GC	DeltaG	DeltaH	DeltaS	TmProd	Sequence
1	20	+	49.0	30.0	.	.	.	.	GACATTGTGAACAGTAAAA
2	21	+	48.4	25.0	.	.	.	.	ACATTGTGAACAGTAAAA



Q.8

```
dna_sequence =
"CTCGGATTTGTAAAGATCATGATCTCATACATAGTACCTAGCCA"

energy_matrix = {
```



```

    'AA': -4, 'AT': -7, 'AC': -5, 'AG': -11,
    'TA': -7, 'TT': -2, 'TC': -3, 'TG': -4,
    'CA': -9, 'CT': -5, 'CC': -6, 'CG': -7,
    'GA': -9, 'GT': -6, 'GC': -4, 'GG': -11
}

total_energy = 0
num_pairs = len(dna_sequence) - 1

for i in range(num_pairs):
    dinucleotide = dna_sequence[i:i+2]
    total_energy += energy_matrix[dinucleotide]

average_energy = total_energy / num_pairs

print(f"Sequence: {dna_sequence}")
print(f"Total base stacking energy: {total_energy}")
print(f"Average base stacking energy:
{average_energy:.4f}")

```

Q.9

```

(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006$ cd Seq2Feature/
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature$ ls
data out_dna __pycache__ q10_input.tsv q9_input.tsv seq2feature_common.py seq_dna2.py
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature$ python3 seq_dna2.py --input_file q9_input.ts
v --property PHY --output_dir BS24B006_results
Completed successfully. Output directory: /home/bioinfo/BS24B006/Seq2Feature/BS24B006_results
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature$ cd BS24B006_results/
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature/BS24B006_results$ ls
avg_Physicochemical.csv
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature/BS24B006_results$ cat avg_Physicochemical.csv

Input_row,Stacking energy,Enthalpy,Entropy,Flexibility_shift,Flexibility_slide,Free energy,Melting Temper
ature,Mobility to bend towards major groove,Mobility to bend towards minor groove,Probability contacting
nucleosome core,Rise stiffness,Roll stiffness,Shift stiffness,Slide stiffness,Tilt stiffness,Twist stiffn
ess
1,1.62,5.44,14.96,2.277,8.697,0.59,43.202,0.988,0.93,6.08,7.027,17.4,0.803,2.395,25.2,23.3
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature/BS24B006_results$ ^C,28.4,18.1
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature/BS24B006_results$ ls
avg_Physicochemical.csv
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature/BS24B006_results$ cd ..
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature$ ls
BS24B006_results data out_dna __pycache__ q10_input.tsv q9_input.tsv seq2feature_common.py seq_dna
2.py
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature$ cat q9_input.tsv
SEQ
ATATATATAT
GCGCGCGCGC
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature$ cut -d',' -f1,8 BS24B006_results/avg_Physico
chemical.csv
Input_row,Melting Temperature
1,43.202
2,97.08
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature$

```

so the output says the melting temprature of

A is 43.202

B is 97.08

which is expected since B is more GC rich

Q.10

```

at: avg nu: No such file or directory
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature/BS24B006_results$ cat avg_Nucleotide.csv
input_row,GC_content,Pyrimidine_TC_content,Purine_AG_content,Keto_GT_content,Adenine_content,Guanine_content,Cytosine_content,Thymine_content
,41.666667,41.666667,58.333333,33.333333,41.666667,16.666667,25.0,16.666667
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature/BS24B006_results$ █

```

AAATGGCCCTAA

- **GC Content:** 41.67% (or 41.666667%)
- **AT Content:** 58.33% (or 58.333333%)